



DELHI PUBLIC SCHOOL NEWTOWN
SESSION 2023–24
HALF TEARLY EXAMINATION

CLASS: X
SUBJECT: MATHEMATICS [SET A]

FULL MARKS: 80
TIME: $2\frac{1}{2}$ HOURS

Answers to this Paper must be written on a paper provided separately.

You will not be allowed to write during the first 15 minutes.

This time is to be spent in reading the question paper.

The time given at the head of this Paper is the time allowed for writing the answers.

Attempt all questions from Section A and any four questions from Section B.

The intended marks for questions or parts of questions are given in brackets [].

This paper consists of 8 printed pages.

SECTION A

*(Attempt **all** questions from this section)*

Question 1

Choose the correct answers to the questions from the given options. (Do not copy the question, write the correct answer only.) **[15]**

(i) The reflection of the point A (7, -4) in the point P (3, -1) is

- (a) (-1, 2) (b) (1, 2) (c) (-1, -2) (d) (1, -2)**

(ii) Which of the following is/are a sequence?

1. $\frac{1}{3}, -\frac{1}{2}, \frac{3}{4}, -\frac{9}{8}$

2. $-\frac{4}{3}, -1, -\frac{2}{3}, -\frac{1}{3}$

3. $\sqrt{2}, \frac{1}{\sqrt{2}}, \frac{1}{2\sqrt{2}}, \frac{1}{3\sqrt{2}}$

- (a) only 1.**
(b) only 2.
(c) only 1. and 2.
(d) all 1., 2. and 3.

(iii) Consider the following information regarding the number of men and women workers in three factories I, II and III.

	Men workers	Women workers
I	30	25
II	25	31
III	27	26

What does the entry in the third row and second column represent if the information is represented as a 3×2 matrix?

- (a) The number of Men workers in factory I.
- (b) The number of Women workers in factory III.
- (c) The number of Women workers in factory II.
- (d) The number of Men workers in factory III.

(iv) The value of x , if on dividing $7y^3 + 2y^2 - xy + 14$ by $y + 3$, leaves a remainder -49 :

- (a) 42
- (b) 37
- (c) 45
- (d) 36

(v) The zeroes of the quadratic polynomial $f(x) = abx^2 + (b^2 - ac)x - bc$ are:

- a) $\frac{b}{ac}$ and $\frac{c}{b}$
- b) $\frac{ab}{c}$ and $\frac{a}{b}$
- c) $\frac{-b}{a}$ and $\frac{c}{b}$
- d) $\frac{b}{a}$ and $\frac{-c}{b}$

(vi) It is given that $\triangle ABC \sim \triangle DFE$, $\angle A = 30^\circ$, $\angle C = 50^\circ$, $AB = 5$ cm, $AC = 8$ cm and $DF = 7.5$ cm. Then, which of the following is true?

- 1. $DE = 12$ cm, $\angle F = 50^\circ$
- 2. $DE = 12$ cm, $\angle F = 100^\circ$
- 3. $EF = 12$ cm, $\angle D = 100^\circ$

- (a) only 1.
- (b) only 2.
- (c) only 1. and 2.
- (d) all 1., 2. and 3.

(vii) If $-1 < 3 - 2x \leq 7$ is expressed as $a \leq x < b$, then the values of a and b are:

- a) $a=2, b=-2$
- b) $a=-2, b=3$
- c) $a=-2, b=2$
- d) $a=-3, b=2$

(viii) If the common difference of an AP is 5, then what is $t_{18} - t_{13}$?

- a) 15
- b) 20
- c) 25
- d) 30

(ix) The table shows the values of a and b , where a is proportional to b .

a	18	36	Q
b	P	54	18

What are the values of P and Q ?

- a) $P=12, Q=27$
- b) $P=27, Q=12$
- c) $P=18, Q=8$
- d) $P=9, Q=3$

(x) Mr. Chanchalani took a health Insurance Policy for his family and paid ₹ 900 as SGST. Find the total annual premium paid by him for this policy, rate of GST being 18%.

- a) ₹ 11800 b) ₹ 10000 c) ₹ 5000 d) ₹ 3600

(xi) In triangle ABC, D is a point on AB. If $\angle ACB = \angle CDA$, AC = 8 cm and AD = 3 cm, the value of BD is:

- (a) $53/3$ cm (b) $55/3$ cm (c) $64/3$ cm (d) $35/7$ cm

(xii) If the points (1, 4), (3, -2) and (p, -5) are collinear, the value of p is:

- a) 4 b) -4 c) 2 d) -2

(xiii) Mr Bikash opened a recurring deposit account in a bank. He deposited ₹ 2500 per month for two years and earned an interest of ₹ 7500. What is the total amount that he received at the time of maturity?

- a) ₹ 67000 b) ₹ 67500 c) ₹ 67550 d) ₹ 68500

(xiv) At what price should I buy a share the value of which is ₹ 100, paying a dividend of 9% so that my yield is 15% ?

- a) ₹ 70 b) ₹ 60 c) ₹ 90 d) ₹ 80

(xv) Assertion (A) : Three unbiased coins are tossed together, then the probability of getting at most 2 tails is $\frac{7}{8}$.

Reason (R) : Favourable number of outcomes do not lie in the sample space of total number of outcomes.

- (a) A is true, R is false
(b) A is false, R is true
(c) both A and R are true
(d) both A and R are false

Question 2

(i) Mr Rastogi has a recurring deposit account in a bank for 4 years at 10% per annum rate of interest. If he gets ₹ 21560 as interest at the time of maturity, Find:

- a) the monthly instalment paid by Mr Rastogi.
b) the amount of maturity of this recurring deposit account. [4]

(ii) If $y = \frac{(a+1)^{\frac{1}{3}} + (a-1)^{\frac{1}{3}}}{(a+1)^{\frac{1}{3}} - (a-1)^{\frac{1}{3}}}$, find the value of $y^3 - 3ay^2 + 3y - a$. [4]

(iii) Given are two Arithmetic Progressions :

1) -8, -5, -2, 1, 4,..... 2) 190, 187, 184, 181,.....

a) Find the sum of the common difference of the two progressions.

b) What is the 31st term of the second progression?

c) Find the sum of the first 20 terms of the first progression.

d) Which term of the two progressions will have the same value?

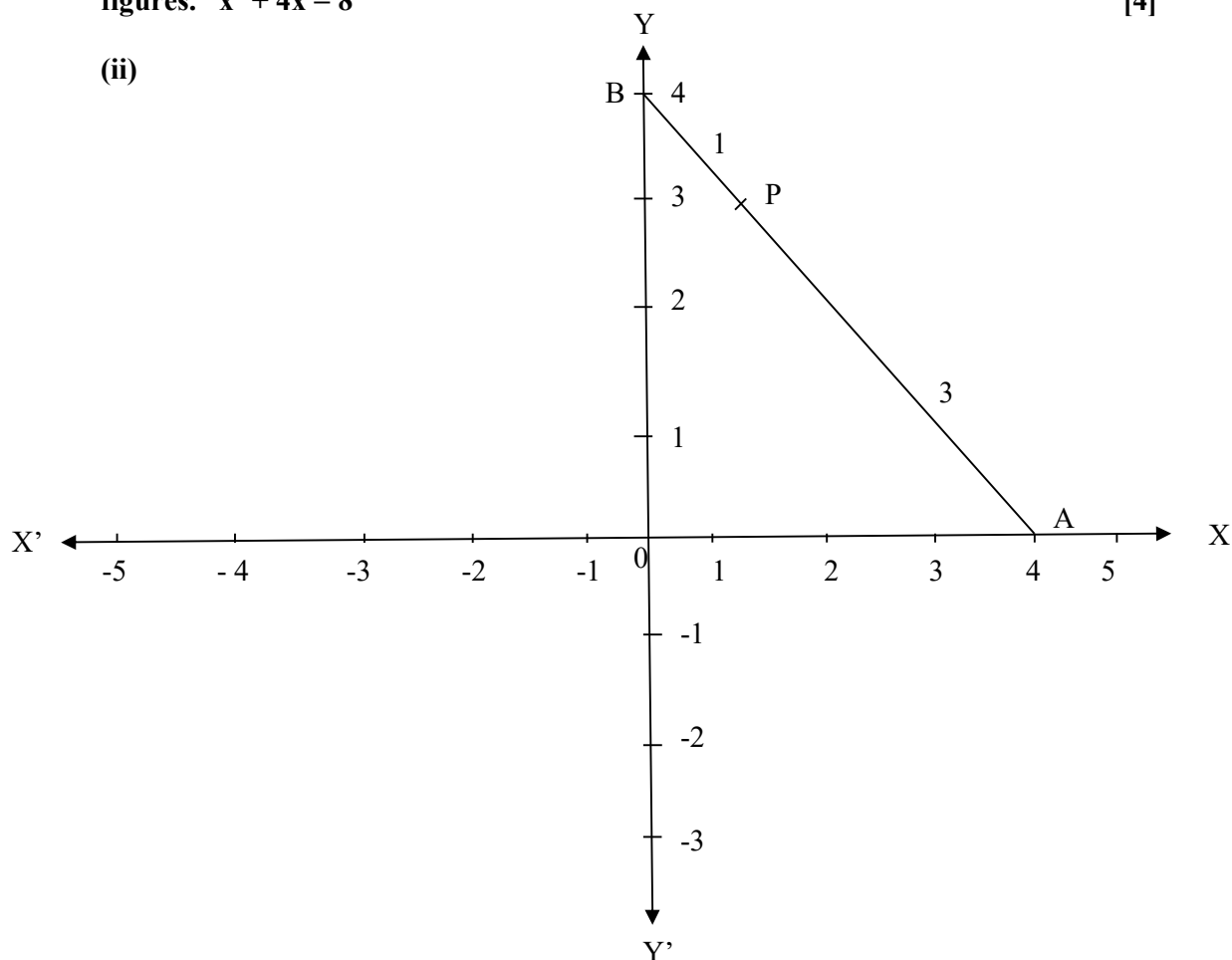
[4]

Question 3

(i) Solve the following equation and give your answer correct to two significant figures. $x^2 + 4x = 8$

[4]

(ii)



In the above figure, A and B are two points on the x-axis and y-axis respectively.

a) Write down the coordinates of A and B.

b) P is a point on AB such that $AP : PB = 3 : 1$. Using section formula find the coordinates of point P.

c) Find the equation of a line passing through P and perpendicular to AB.

[4]

(iii) Use graph sheet to answer this question. (Take 2 cm = 1 unit along both x and y axes).

a) Plot A, B, C where A(0, 4), B(1, 1) and C(4, 0)

b) Reflect A and B on the x axis and name them as E and D respectively.

c) Reflect B through the origin and name it F. Write down the coordinates of F.

- d) Reflect B and C on the y axis and name them as H and G respectively.
 e) Join points A, B, C, D, E, F, G, H and A in order and name the closed figure. [5]

SECTION B

(Attempt any four questions from this section)

Question 4

(i) Two dice, one blue and one grey, are thrown at the same time. Write down all the possible outcomes. What is the probability that the sum of the two numbers appearing on the top of the dice is

- a) 8 ?
 b) less than or equal to 12 ? [3]

(ii) If $A = \begin{bmatrix} 3 & 4 \\ 2 & -1 \end{bmatrix}$, find a matrix X satisfying $A^2 = A + 2X$. [3]

(iii) The polynomials $ax^3 + 3x^2 - 3$ and $2x^3 - 5x + a$ when divided by $(x - 4)$ leave the remainders R_1 and R_2 respectively. Find the values of “a” in each of the following cases, if

- a) $R_1 = R_2$
 b) $R_1 + R_2 = 0$
 c) $2R_1 - R_2 = 0$ [4]

Question 5

(i) Solve the following inequation and represent your solution set on the real number line: $-5(x-9) \geq 17-9x > x+2$, $x \in \mathbb{R}$ [3]

(ii) The first term of a G.P is 1. The sum of the third and fifth terms is 90. Find the common ratio of the G.P. [3]

(iii) A company with 10,000 shares of ₹ 50 each, declares an annual dividend of 5%.

- a) What is the total amount of dividend paid by the company?
 b) What would be the annual income of a man who has 72 shares in the company?
 c) If he receives only 4% on his investment, find the price he paid for each share. [4]

Question 6

(i) What number should be added to $2x^3 - 3x^2 - 8x + 2$ so that the resulting polynomial leaves a remainder 10 when divided by $2x + 1$. [3]

(ii) If the mid-point of the line joining (3, 4) and (k, 7) is (x, y) and $2x + 2y + 1 = 0$, find the value of k. [3]

(iii) Mrs Basu went to a departmental store and bought the following items. The discount and GST rates and the quantity of each items and marked price of each are given below:

Item	Quantity	Rate per item(₹)	Discount	GST
Dinner Set	1	₹ 3,000	10%	18%
Muesli	3	₹ 250	5%	NIL
Bakery products(unpacked)	2	₹ 175	NIL	NIL

Calculate the:

a) The total amount of GST paid.

b) The total bill amount paid by Mrs Basu. [4]

Question 7

(i) If $x = 2$ and $x = 3$ are roots of the equation $3x^2 - 2kx + 2m = 0$, find the value of k and m. [3]

(ii) On a map drawn to a scale of 1 : 50,000, a rectangular plot of land PQRS has the following dimensions. PQ = 6cm; QR = 8cm and all the angles are right angles. Find:

a) the actual length of the diagonal distance PR of the plot in km.

b) the actual area of the plot in square km. [3]

(iii) If $\frac{x}{(b-c)(b+c-2a)} = \frac{y}{(c-a)(c+a-2b)} = \frac{z}{(a-b)(a+b-2c)}$, prove that $x+y+z=0$ [4]

Question 8

(i) A box contains a certain number of balls. On each of 60% balls, letter A is marked. On each of 30% balls, letter B is marked and on each of the remaining balls, letter C is marked. A ball is drawn from the box at random. Find the probability that the ball drawn is marked

a) C

b) A or B

c) neither B nor C [3]

(ii) Find a matrix A satisfying the equation: $A \begin{bmatrix} 2 & 3 \\ 4 & 5 \end{bmatrix} = \begin{bmatrix} 0 & -4 \\ 10 & 3 \end{bmatrix}$ [3]

(iii) The coordinates of the vertex A of a square ABCD are (1, 2) and the equation of the diagonal BD is $x + 2y = 10$. Find:

- a) the equation of diagonal AC
- b) the coordinates of the point of intersection of the diagonals.
- c) the coordinates of the intercepts made by the diagonal AC on the coordinate axes.

Does the diagonal AC pass through the origin?

[4]

Question 9

(i) Without using graph papers, answer the following question.

The point (3, -2) on reflection in the x-axis is mapped as A' and the point B on reflection in the y-axis is mapped onto B' (-4, 3)

a) Write down the coordinates of A' and B.

b) Find the slope of the line A'B, hence find its inclination with the positive direction of the x-axis.

[3]

(ii) Divide ₹ 35400 into two parts such that if one part is invested in 9%, ₹ 100 shares at 4% discount and the other in 12%, ₹ 50 shares at 8% premium, the annual incomes are equal.

[3]

(iii) A cottage industry produces a certain number of toys in a day. The cost of production of each toy (in rupees) was found to be 55 minus the number of articles produced in a day. On a particular day, the total cost of production was ₹ 750. If x denotes the number of toys, form the quadratic equation to find x .

[4]

Question 10

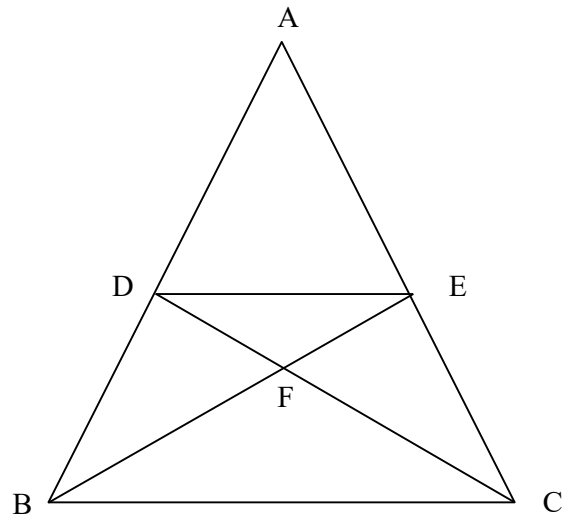
(i) The 4th, 6th and the last term of a geometric progression are 10, 40 and 640 respectively. If the common ratio is positive, find the first term, common ratio and the number of terms of the series.

[3]

(ii) If $A = \begin{bmatrix} 3 & 1 \\ -1 & 2 \end{bmatrix}$, show that $A^2 - 5A + 7I_2 = 0$ where I is a unit matrix of order 2.

[3]

(iii)



In the above figure, $DE \parallel BC$ and $AD : DB = 5 : 4$. Find:

a) $DE : BC$

b) $DF : DC$

c) $\frac{\text{Area}(\triangle DFE)}{\text{Area}(\triangle CFB)}$

d) $\frac{\text{Area}(\triangle DEF)}{\text{Area}(\triangle DCE)}$

[4]